Hand-transmitted Vibration System for Measuring Vibration during Drilling Operation

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ABSTRACT

Vibration exposure affects every organ in the human body differently, leading to resistance and potential health risks. Hand-transmitted vibration (HTV) is an occupational hazard in industries, causing hand-arm vibration syndrome (HAVS) and vascular, neurological, and musculoskeletal symptoms. Regular monitoring is crucial to prevent these effects. Therefore, a wireless sensor module is required to monitor the individuals for risk continuously. This study develops an indigenous vibration monitoring device using a micro-electro-mechanical systems (MEMS) accelerometer that can be worn on the individual to measure their vibration exposure. In this study, the effect of using different glove materials was analyzed keeping the posture, feed rate and the drill material constant. The HTV exposures of ten participants were measured under different conditions (barehand, wearing nylon, rubber, and leather gloves) while drilling teak wood using an impact drill. The total frequency-weighted rms acceleration magnitude (a_{hv}) while wearing nylon, rubber, and leather gloves was reduced by 16.78%, 30.91% and 23.78% when compared to barehand. The A(8) or exposure action value (EAV) has a reduction effect when gloves are used. The reduction was 12.84%, 35.85%, and 22.36% for nylon, rubber and leather when compared with barehand respectively.

The most significant vibration amplitudes occurred at frequencies of 100 Hz and 1000 Hz. The study showed that using rubber gloves reduces the vibration exposure action value and can effectively improve the efficiency of the drill operator. Thus, the findings emphasize the importance of selecting appropriate personal protective equipment (PPE) to enhance worker safety and comfort in environments with significant vibration exposure.

Keywords: Daily exposure value, frequency-weighted rms acceleration, hand-transmitted vibration, hand-arm vibration syndrome, vibration magnitude.

INTRODUCTION

Hand-transmitted vibration (HTV) refers to the mechanical shock or mechanical vibration directly transferred to the hand while using a handheld vibration tool or contacting vibrating workpieces in production(Fu et al., 2022). Globally, millions of workers are exposed to hand and arm vibrations while using heavy power tools. Various occupational factors have led to the development of hand-pain-related disorders. Paraesthesia, tingling in the fingers, diminished sense of touch, and insomnia are some of the complications associated with HTV (Forouharmajd et al., 2016). The International standard organization (ISO) -5439-1,2 set a standard to measures, evaluates, and assesses HTV using four characteristics: magnitude, exposure time, frequency, and direction. Root-mean-square acceleration quantifies vibration magnitude near the power tool gripping zone. Daily exposure duration and frequency weighting acceleration are determined through sensors. Daily vibration dose is calculated by combining frequency weighting acceleration and exposure duration. According to ISO-5439, several studies (Dong et al. 2021) have been conducted to assess HTV and reduce exposure to vibration by developing wearable sensors for monitoring vibration. The Health Effects Laboratory Division of the National Institute for Occupational Safety and Health has conducted 20 years of research on biodynamic responses, HTV exposure, and biological effects on individuals exposed to vibration environment. Hand-Arm Vibration Syndrome

(HAVS) is a group of vascular, musculoskeletal, and neurological symptoms that can result from extended use of vibrating hand-held power equipment. The HTV may affect one's perception, fine motor skills, and performance. In the long term, it can cause nerve function disorders, muscular tissue changes, circulation disorders, and bone and joint damage.

There are various methodologies to quantify the vibration transmitted to the body. Accelerometers have been the most widely utilized form of transducers because they can convert a surface's acceleration into an electrical signal in one way or another (Griffin 1997). Initially, a frequency modulation (FM) tape recorder was connected to the accelerometers via preamplifiers. The FM tape recorder was also used to record the push, grip and dynamic forces as measured by the handle (Kihlberg 1995). Later, piezoelectric and MEMS based accelerometers are involved in detecting vibration (Welcome et al. 2015). Many studies had investigated HTV from industrial workers, drummers, cyclists and many more whose are exposed in vibration environment (Chiementin et al. 2013; Roseiro et al. 2018a; Clemm et al. 2021). These studies measure the exposure of vibration and vibration magnitude of the tools used in each environment.

To reduce the vibration, several studies have been done. A study was conducted to create a low-vibration impact machine that is easy for users to operate, incorporating Auto-Tuning Vibration Absorber (ATVA) and integrated vibration isolation. Several methods exist for managing handle vibration, such as utilizing anti-vibration gloves to separate the hand from the vibrating handle (Ko et al. 2011). Minimizing the risks of vibration-induced injuries can involve employing various methods, such as utilizing personal protective equipment. Gloves have been shown to have both beneficial and detrimental effects on hand exertion. Additionally, the vibration energy that passes to and is taken in by the hand and arm system leads to the relative compression and extension of tissues (Singh and Khan 2014). Gloves cause a slight rise in the transmission of vibrations to the wrist and forearm at lower frequencies. The effectiveness of different

gloves based on material contributes to reducing the HTV through hand-held power tools is predicted using the transmissibility model of a bent hand-arm system (Sabu et al. 2022).

Thus, the calculation of exposure is based on parameters for vibrating equipment and the estimated exposure time. The existing approach of monitoring HTV has a larger investment. However, these systems often present certain limitations, especially in specific applications where enhanced flexibility, real-time monitoring, and cost-effectiveness are crucial. Based on the limitations of the existing systems, this study focuses on developing an indigenous system that is portable and capable of accessing and recording data, allowing many users to access it. The system meets the requirements of a real-time recording and data logger, has a lower number of components, lightweight, adaptable, and low-cost. This study involves the comparative study of vibration exposures obtained with bare hands and with various gloves made of various materials using the developed vibration system.

METHODS

Experimental setup

The experimental setup comprises of ESP32 microcontroller interfaced with ADXL345 through I2C communication. The ESP32 is a powerful, low-cost microcontroller with integrated Wi-Fi and Bluetooth capabilities. It supports I2C (Inter-Integrated Circuit) communication, which allows easy interfacing with the ADXL345 accelerometer. The integrated Wi-Fi module allows the ESP32 to connect to wireless networks, facilitating the wireless transmission of data to a remote server or a cloud.

The ADXL345 accelerometer is a compact, low-power, 3-axis accelerometer that provides high-resolution measurement. It is capable of measuring acceleration within the $\pm$2 g, $\pm$4 g, $\pm$8 g, and $\pm$16 g range, covering a wide spectrum of vibration intensities. In this experiment, we have considered $\pm$16 g resolution, which indicates that the sensor can accurately measure along each axis. The $\pm$16 g range of the ADXL345

makes it appropriate for orientation detection, vibration monitoring, and gesture recognition. It can measure both small and large accelerations as per requirement.

Data rate is another parameter that is taken into consideration. It allows the frequency of data output from the sensor to be controlled. A high data rate means the sensor samples acceleration data more frequently, providing more data points per unit time. It also affects the resolution of the acceleration data captured by the sensor, allowing more detailed information about changes in acceleration over time. The sensor has data output rates of 0.1 Hz to 3200 Hz, which helps capture data at different frequencies. It is related to the bandwidth of the accelerometer. The data rate setting influences the maximum frequency of acceleration changes that the sensor can capture. The data rate chosen impacts the power consumption of the sensor. Higher data rates generally require more power as the sensor is sampling and processing data more frequently. The clock frequency plays a major role in data transfer between the ESP32 module and the ADXL345 accelerometer. The clock frequency of 3.4 MHz in ESP32 is used for data transfer, which influences the speed at which the ESP32 microcontroller can communicate with external devices, such as the ADXL345 accelerometer. A higher clock frequency allows for faster data transfer speeds during sensor readings. It gives optimal performance and ensures that data is transferred accurately and efficiently between the microcontroller and the sensor.

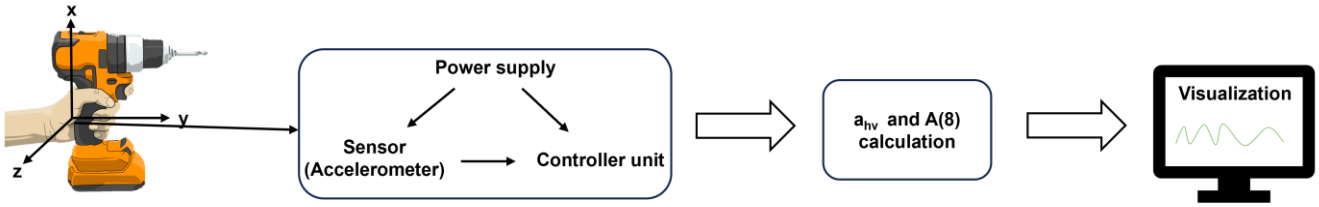


Figure 1. The schematic diagram of the vibration measuring system

The system begins with the controller unit, which consists of an ESP32 microcontroller operating on battery power. ADXL345 accelerometer is interfaced with the control unit, which is mounted on a drill to capture vibration data. The collected data is then wirelessly transmitted to a remote server. The data is acquired from the individuals who are working on power tools through the wearable sensors, then the data is processed to calculate the frequency-weighting rms value (a_{hv}) and daily exposure value (A(8)) through MATLAB (R2023 b). The schematic represents the data acquisition process and how it is stored for the future (Figure 1). The results of this analysis are used to assess **v**ibration exposure, providing insights into the impact of vibrations on the operator's hand. Concurrently, the data is available for real-time visualization during drilling, providing immediate feedback on vibration patterns. This end-to-end process ensures continuous monitoring, real-time visualization, and thorough assessment of vibrations during drilling operations, enhancing safety and ergonomics.

Analysis of vibration exposure

The developed vibration measuring system evaluates vibration exposure by considering multiple parameters, including acceleration measurement, transducer placement, and measurement duration. The ISO standards define the methods and parameters that impact this evaluation process. The transducers must be placed between the handle grip and the hand for accurate HTV calculations. In this study, the vibration accelerations were measured as follows in the three orthogonal coordinate system directions: z-axis, which is aligned along the hand's second metacarpus bone; y-axis parallel to the grip's longitudinal axis; x-axis perpendicular to the z-axis (both of these axes are normal to the grip's longitudinal axis) (Figure 2). The angle of the metacarpal bone is 45º to the vertical when the hand grasps the grip [ISO 5349-1,2001].

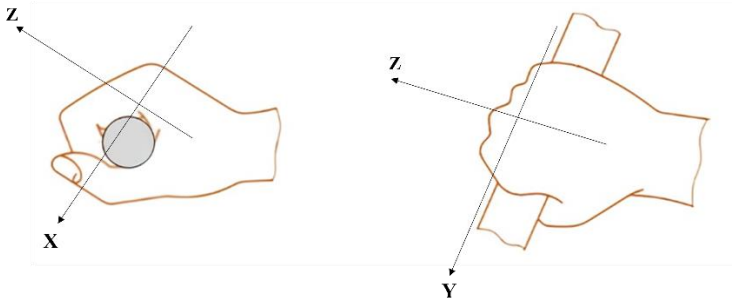


Figure 2. Coordinate system for the hand

The band-limiting filters for the vibration measurements are set to 6.3 Hz and 1258.9 Hz [(Reference number ISO 8041:2005(E) ISO 8041 Second edition Human response to vibration-Measuring instrumentation 2005). The frequency weighting functions for the HTV evaluation is the product of the band-limiting filter and frequency weighting filter.

$$H(s) = H_b(s) . H_w(s) \quad (3)$$

To evaluate the exposure, the frequency-weighted rms values in the triaxial directions (a_{hwx}, a_{hwy}, a_{hwz}) are considered. Mathematically, exposure evaluation is calculated as the square root of the sum of squares of the three component values.

$$a_{hv} = \sqrt{a_{hwx}^2 + a_{hwy}^2 + a_{hwz}^2} \quad (4)$$

Exposure action value (EAV) or A(8) for different durations is calculated as

$$A(8) = a_{hv}\sqrt{\frac{T}{T_0}} \quad (5)$$

where, T is the daily exposure time and T_0 is the reference duration 8 h (28,800 s). If the exposure is such that it has different vibration magnitudes, A(8) is calculated as,

$$A(8) = \sqrt[2]{\frac{1}{T_0}\sum_{i=1}^{i=n} a_{hvi}T_i} \quad (6)$$

where,

i = different vibration magnitudes

The employer must control exposure levels at 2.5 m/s^2 (A(8) threshold) for the hand-arm system, and workers must not be exposed to levels exceeding 5 m/s^2 for the hand-arm system. ISO 10819 defines two ranges of frequency for assessing the effectiveness of vibration-damping gloves, which include low (25–200 Hz) and medium frequency (200–1250 Hz). In this study, to assess the frequency changes of the measured data these ranges were evaluated [ISO 10819]. According to ISO 5349-1, the frequency weight filter having frequency values above 1250 Hz is much less important than the low-frequency range, so the frequency ranges below 1250 Hz were analyzed in the study's findings.

Procedure

A pilot study of ten participants using the developed sensor module was used to measure the vibrational exposure transmitted by using a hand-held drilling machine (impact drill). According to ISO 8662-3 standard, the sensor is positioned where the hand tool and hand come into contact for capturing the vibrations, and the controlling unit is attached to the hand using a Velcro band. A teak wooden block is used to investigate the effect of vibration while drilling. An impact drill machine (Johnson JN-ED-101) was used in the study with a drilling bit of size (6 mm), which eliminates the effect of the drill bit size on the vibration transmission. The impact drill machine used has 400 W power and 2600 rpm. A constant feed of 5 mm/min is considered in the experiment. During the experiment, a clamp was used to hold the block in order to reduce the unwanted movement that could cause artifacts in the data.

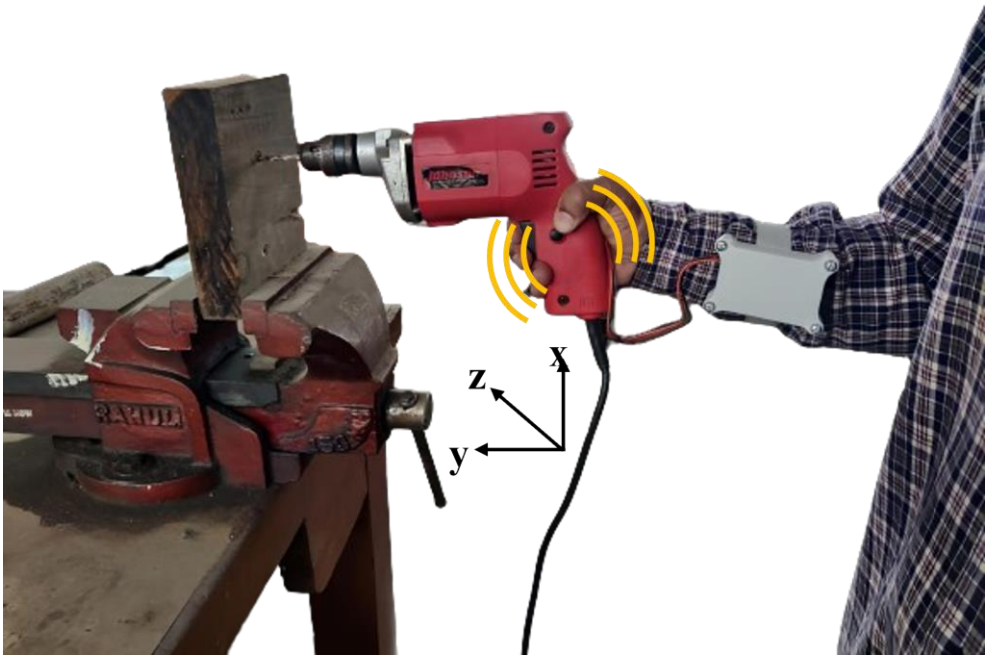


Fig.3. Experimental setup of the drilling operation

Ten healthy male subjects (right-handed) between the age group of 20 to 25 (21.5 ± 1.67) were considered for the study. The vibrations were measured for 5 min of each condition with 5 trials. The subjects were informed about the experiment and approved by the Institute's ethical clearance (number). The subjects are made to understand the safety measures of work and handling the impact drill machine with some trail measurements. This study involves the comparative study of vibration exposures obtained with bare hands and with various gloves made of various materials using the developed vibration system.

Table 1. Different glove materials used for evaluating vibration exposure

S.No	Condition	Glove Materials
1	C1	Control (Barehand)
2	C2	Nylon

3	C3	Rubber
4	C4	Leather

The vibration exposures were measured for the four conditions, which are listed in Table 1, and using one-way ANOVA with Tukey post-hoc test was used to find the
statistically significant differences among the vibration exposures. The statistical analysis was performed using the IBM SPSS software (version 24, Chicago, Illinois, USA).

Results

The developed vibration measuring system

The indigenous wireless, battery-operated vibration measuring system was developed using an ESP32 microcontroller integrated with an ADXL345 accelerometer through I2C communication. The ESP32 allows for wireless data transmission and battery management, ensuring portability and convenience. The ADXL345 accelerometer, connected to the ESP32 via I2C, is configured to capture vibration data at a high rate which runs at a clock frequency of 3.4 MHz, facilitating fast and efficient data transfer. The sensor setup is powered by a rechargeable battery with an integrated charging module, ensuring continuous operation. The ESP32 is programmed to read acceleration data from the ADXL345 and store it in CSV format, making it easy to analyze the vibration data from each axis (x, y, z). The real-time can also be visualized for better analysis. By enabling Wi-Fi, the collected data can be transmitted wirelessly to a designated server or cloud storage. The result is a compact, efficient, and portable vibration sensor capable of capturing high-frequency vibration data for various applications.

The printed circuit board (PCB) is designed in EasyEDA (v6.4.31, Shenzhen, China) software. The PCB is designed for a compact, lightweight controller unit (Figure 1c). The casing for the controlling unit of the vibration sensor is designed with careful consideration of component placement. It is designed using CAD (SolidWorks 2019, USA) and fabricated using polylactic acid (PLA) via a 3D printing mechanism. It includes the ESP32 and charger module PCB board, ensuring they remain cool during operation (bottom part). A compartment (middle part) is made above the PCB board for the placement of the battery, preventing any heat buildup so that it does not affect the PCB board. The design incorporates a space for an on/off switch, allowing easy control of the device. This design ensures compatibility, battery charging and user-friendly casing for the vibration sensor-controlling unit (Figure 1a). The sensor casing is also designed using CAD (SolidWorks 2019, USA) and fabricated from thermoplastic polyurethane (TPU) material using 3D printing technology. This casing is ergonomically designed to ensure that it does not cause any discomfort or poke the hand while holding the impact drill machine, providing a comfortable grip for drilling operations. The design carefully considers the placement and protection of wire connections and sensor components, ensuring that the functionality and sensitivity of the sensors are maintained. This integration of material and design principles results in a robust, user-friendly casing that enhances the overall drilling experience (Figure 1b).

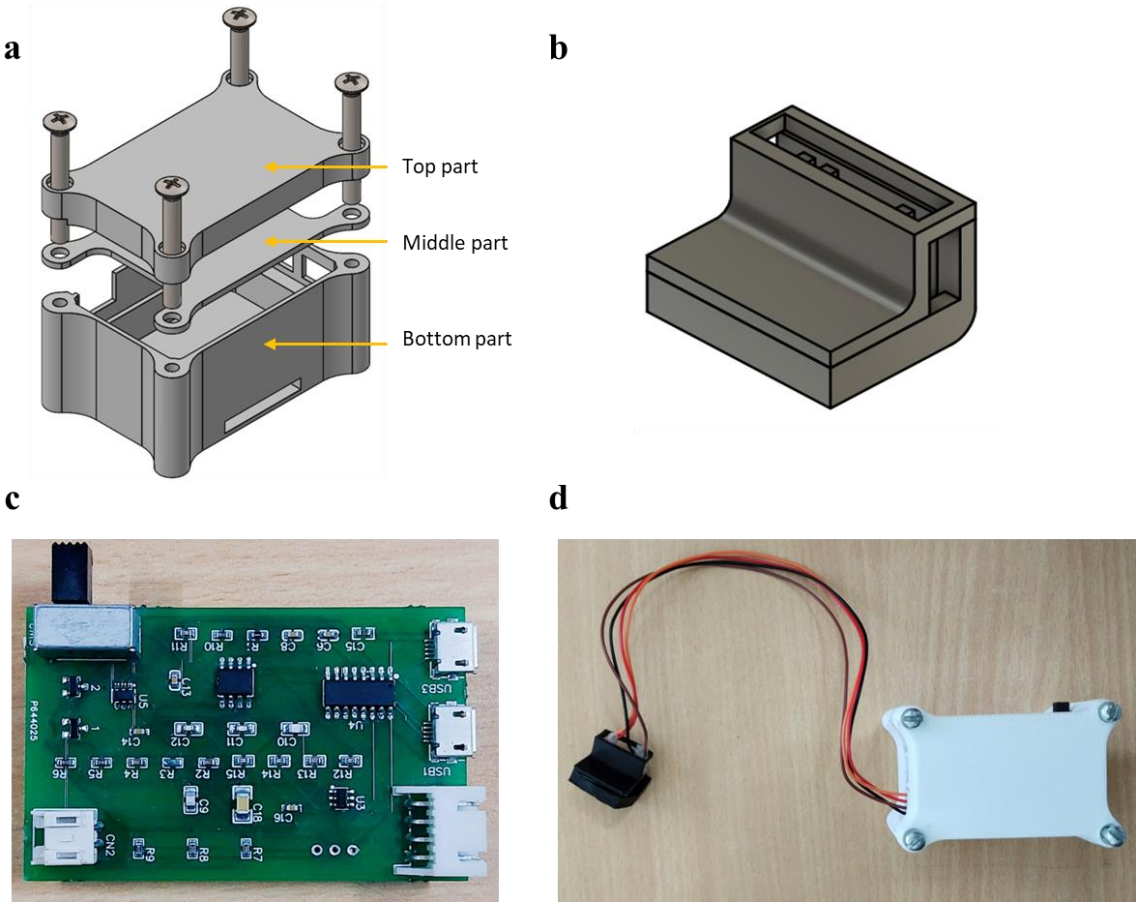


Figure 1. The CAD model of (a) sensor casing (b) controller unit casing and (c) PCB of the indigenously developed vibration measurement system (d) The complete assembled vibration system.

To validate the vibration measuring system that uses an ESP32 microcontroller and an ADXL345 accelerometer, a systematic approach was employed. The objective was to ensure accurate measurements by using a smartphone's vibration as a reference. Initially, the hardware setup involved connecting the ADXL345 to the ESP32 with proper wiring for power, ground, and I2C communication. The accelerometer was placed in direct contact with a smartphone. The vibration was activated, and data was collected from the ADXL345 over a specified period, such as 30 seconds. This process was repeated for varying vibration intensities to gather a comprehensive dataset. The recorded accelerometer data was

analyzed to establish a correlation between the phone's known vibration intensity and the measured acceleration values. It is inferred that the device was able to capture the low vibration magnitudes.

Evaluation of frequency-weighted rms value

The comparison between frequency-weighted rms values in the x, y, and z directions of the vibration magnitude was observed for volunteers without using gloves for bare hand (Figure.5). During the drilling operations, there was no dominant axis recorded for HTV exposures, although other professions had the dominant axis. For example, for professional drummers, the y-axis was dominant (Roseiro et al. 2018b). The average frequency-weighted rms value of HTV exposures for the x, y, and z axes was 7.41 m/s^2 (6–8 m/s^2), 7.17 m/s^2 (4–13 m/s^2), 6.28 m/s^2 (4–8 m/s^2) respectively. Participants whose frequency-weighted rms values in the y-axis are higher might have a different posture or hand position while performing the task, leading to higher vibrations along the y-axis. In actual situation, while using the impact drill machine, a high vibration magnitude is produced. The participants had to stop and restart the vibration; this leads to producing a high vibration magnitude during the initiation time. The HTV exposure depends on the type of material used while drilling. There are other hand-held tool operations that produced higher frequency-weighted HTV magnitudes. For example, in pneumatic drill operators,
the vertical vibration's weighted magnitude was found to be 18.01 m/s^2, whereas in hydraulic drill operators, it was 26.32 m/s^2 (Phillips et al. 2007). Studies revealed that for other heavy power tool operations, the frequency-weighted magnitude was reported to be 30.71 m/s^2 for the rock drill, 39.14 m/s^2 for the pneumatic rock drill (Rempel et al. 2019) and 24.81 m/s^2 for light stone hammers (Bovenzi et al. 1994). In the present study, a softer material (teak wood) block was used for drilling operations. The frequency-weighted vibration magnitude (a_{hv}) was found to be 12.33 m/s^2 for teak wood without using any gloves.

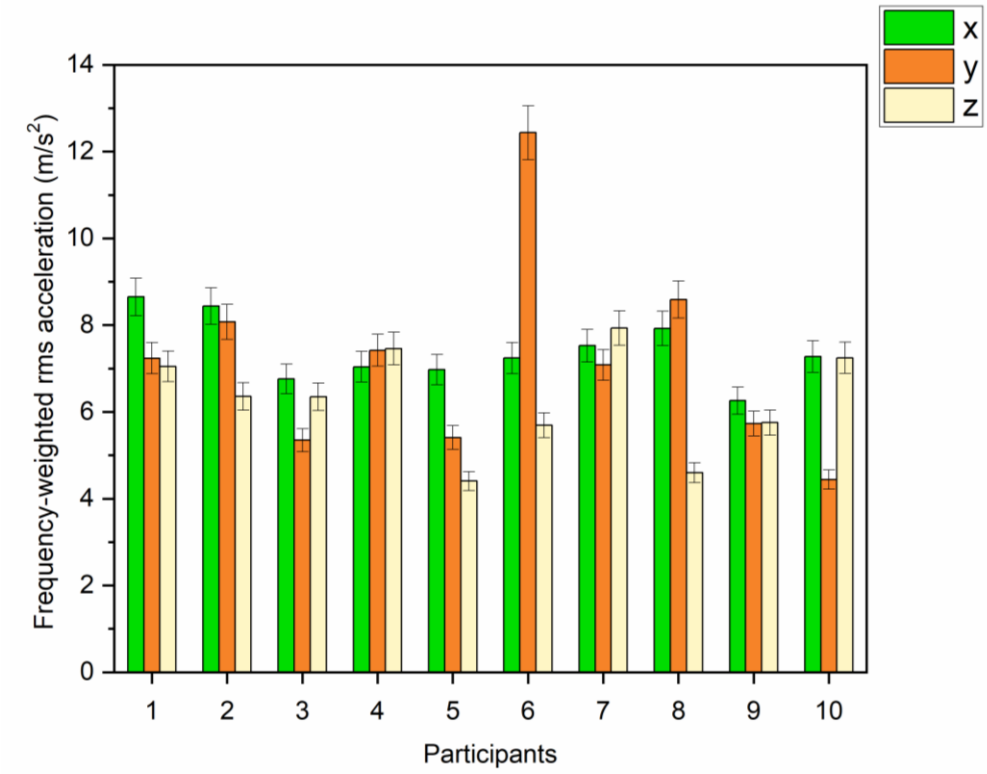


Figure 5. Frequency-weighted rms vibration value of the participant in the three directions (barehand)

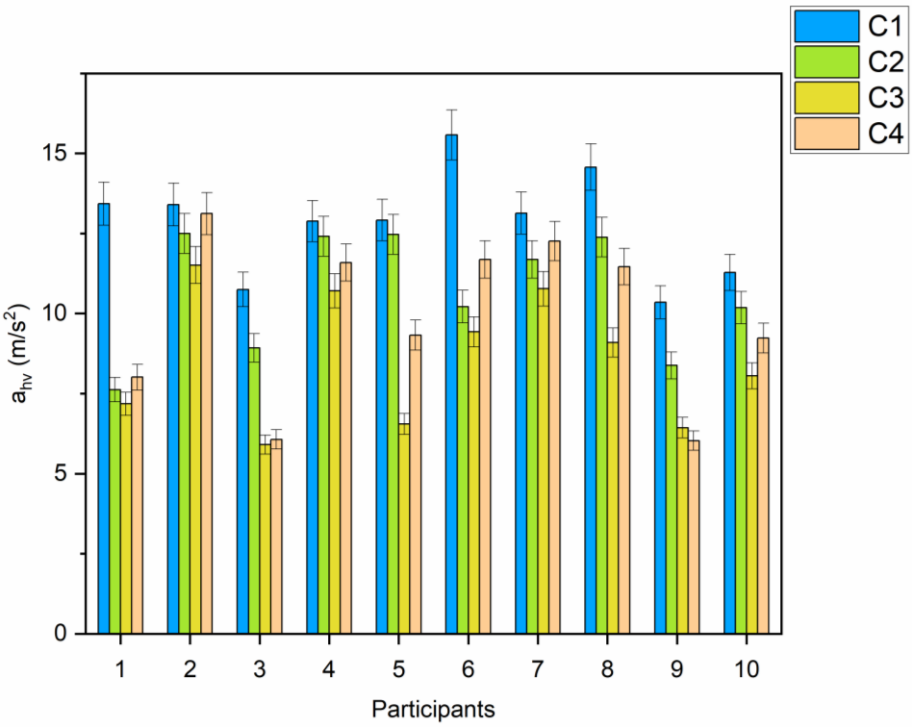


Figure 6. Frequency-weighted vibration magnitude of the individuals for the four conditions

During the drilling operation, there was a reduction in frequency weighted vibration magnitude (a_{hv}) for every participant using gloves for all conditions C2, C3 and C4 (Figure 6). The statistical data was analyzed and inferred that C1 had a significant difference compared to C2 ($p < 0.05$). Similarly, C1 also showed a significantly higher compared to C3 ($p < 0.05$). The difference between C1 and C4 was marginally significant ($p < 0.05$). In addition, C2 demonstrated significantly higher than C3 ($p < 0.05$). Although C2 had a higher mean compared to C4, this difference was not statistically significant. Conversely, C3 exhibited a significantly lower compared to C4. These findings suggest significant effects, with C1 generally exhibiting higher levels compared to other conditions.

Evaluation of A(8)

To prevent individuals/operators from HAVS, HTV exposures of impact drill machines should be reduced. The HTV exposure cannot be reduced only by reducing the work hours per day. Wearing gloves can be an alternate method to minimize the transmission of vibration to the palm (Md Rezali and Griffin 2016).

The average A(8) or the EAV values of 10 participants were observed in all the glove conditions; the A(8) value was lower than the bare hand (Figure 7). For the 5-minute exposure duration, the A(8) values across different conditions—barehand, rubber, nylon, and leather reveal notable variations in risk levels. It is found that the triggering time to reach EAV is <15 mins (Lai et al. 2019). For barehand use, the A(8) value reaches up to the maximum of 1.46 m/s^2. This implies a high risk of vibration-induced health issues, necessitating immediate action to reduce exposure. Nylon and leather gloves offer some reduction in vibration exposure compared to barehand use. Rubber gloves typically provide a further reduction in vibration exposure, potentially lowering the A(8) value below the EAV, indicating minimal risk of vibration-related health problems, although vigilance and routine monitoring are still required.

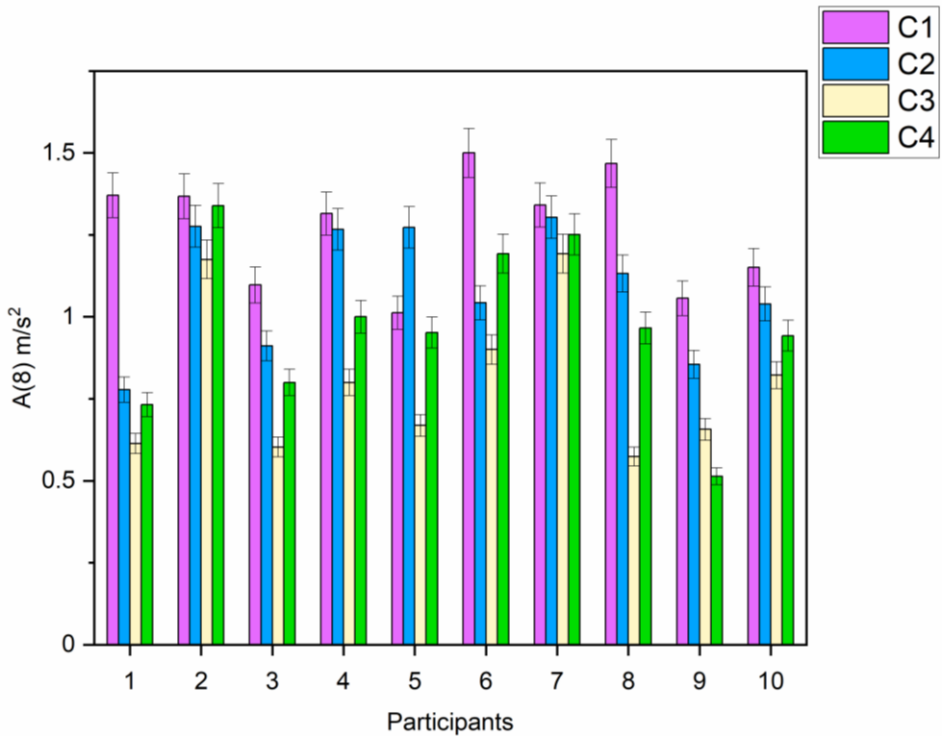


Fig.7. Daily exposure value of the individuals for the four conditions

The one-way ANOVA post hoc analysis for the dependent variable A(8) was conducted to compare the mean differences among four conditions: barehand, nylon, rubber, and leather. The results indicate significant differences between several pairs of conditions. The mean difference between C1 and C2 was

less, with a significance level of $p < 0.001$ and with a confidence interval (CI) of 95%, indicating that barehand has a significant difference from rubber. Specifically, the C1 and C3 had a significance level of $p < 0.001$, indicating that barehand has a significant difference from rubber. Similarly, the comparison between C1) and leather (C4) revealed a significant mean difference ($p = 0.022$), showing that barehand measurements are also significantly lower than those for leather. Furthermore, the significance level between C2 and C3 $p < 0.024$, indicating that nylon measurements are significantly lower than those for rubber. These findings highlight that rubber and leather conditions have higher significant differences compared to barehand and nylon conditions for the dependent variable A(8), while the differences between the other pairs were not statistically significant.

Frequency analysis

The frequency spectrum in one-third octave band was analyzed based on the performance done by the four conditions. The results obtained indicated that the dominant frequencies contributed to the vibrations that were experienced on the bare hand and three types of gloves. The most significant vibration amplitudes occur at frequencies of 100 and 1000 Hz (Figure 8). According to the ISO 5439 standards, prolonged exposure of hands at the frequency range of 10 Hz might cause severe physical discomfort and potential injuries. If the hands are exposed for a longer time, the standards suggest that the person will be affected by Hand-Arm Vibration Syndrome (HAVS). Compared to the frequency range of 10 Hz, the impacts by frequency ranges at 100 and 1000 Hz would be lesser. These records help the workers to know the importance of getting affected by specific frequencies and save them from various health hazards. Further study is required on the methods to reduce the harmful frequency vibration range to prevent the workers from health hazards. The resonant frequency of fingers ranges from 150- 300 Hz (Dong et al. 2004) and the vibration is transmitted only to the finger and hand when the frequency is greater than 100 Hz. the

vibration from the impact drill is predominantly transferred to the finger and hand where the peak is at 250 Hz (Figure 8).

The vibration-damping capacities of various materials can be seen in the frequency spectra of vibrations transmitted by hand under various conditions (barehand, nylon glove, rubber glove, and leather glove). These variations are substantial. Using the bare hand as a reference, the spectrum shows clear peaks at particular frequencies, signifying the natural vibration frequencies that are transferred to the hand in the absence of any dampening substance. For the barehand, rms acceleration remains relatively low at frequencies below 100 Hz, with values around 1-2 m/s^2. However, as the frequency increases, there is a noticeable rise in rms acceleration, with peaks around 100 Hz reaching approximately 1.3 to 2.5 m/s^2 and a sharp increase near 1000 Hz, where values peak up around 4.5 to 5.5 m/s^2. This indicates significant vibration exposure at higher frequencies when no glove is worn. The nylon glove condition displayed similar peak frequencies but with reduced magnitudes. At 100 Hz, the peak reached about 3 m/s^2, while near 1000 Hz, the peak was reduced to around 5 m/s^2. The frequency spectrum of rubber gloves shows the lowest peak amplitudes, indicating the greatest damping effect, maintaining low rms acceleration values around 1.2 m/s^2 at lower frequencies and significantly reducing peaks around 100 Hz and 1000 Hz to approximately 2 to 2.8 m/s. Leather gloves show effective damping at lower and mid frequencies, with rms acceleration values around 0.5 to 1.5 m/s^2, but higher frequencies still exhibit prominent peaks, particularly in the y-axis, where values reach up to 3 m/s^2. This suggests that leather gloves offer a moderate vibration damping between protection and dexterity.

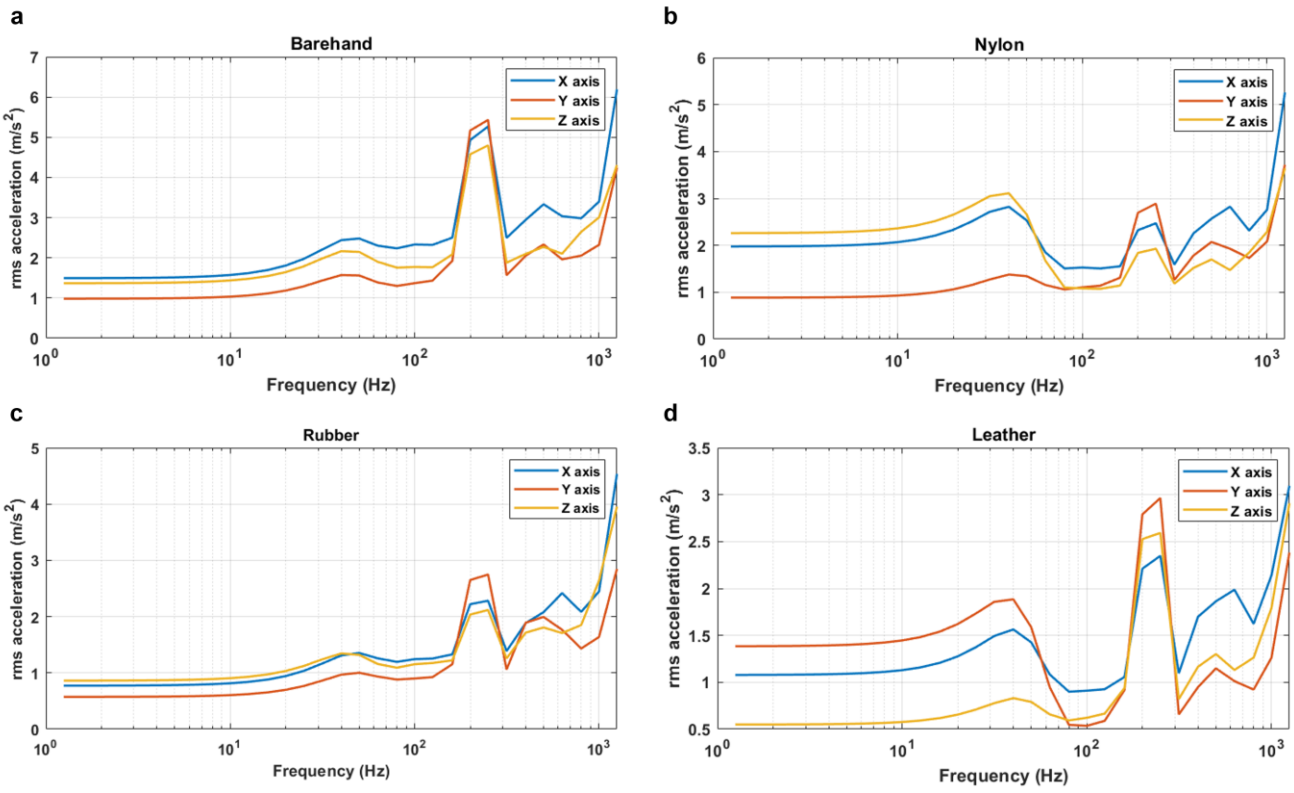


Fig.8. Frequency spectrum at 1/3 octave band for the four conditions a) barehand b) nylon c) rubber d) leather

Discussion

The findings of this study clearly indicate the significant role of gloves in mitigating vibration exposure during drilling operations. The reduction in frequency-weighted vibration magnitude (a_{hv}), A(8) and frequency analysis highlights the protective benefits of using gloves, particularly rubber gloves.

Nylon gloves are suitable because they reduce vibration significantly and offer some protection without significantly compromising dexterity. Rubber gloves provide the greatest damping effect, making them ideal for high-frequency vibration environments or other situations where the highest level of protection is needed. Leather gloves offer a balanced approach to maintaining dexterity and comfort; they are more flexible and comfortable than nylon but less so than rubber. However, for participant 9, the average value

while using leather gloves (C4 condition) was found to be lesser than using a rubber glove (C3 condition). This might be due to a slight increase in the gripping force of participant 9 on the tool, which resulted in a decrease in the a_{hv} value while wearing a leather glove (Lindenmann et al. 2021). The glove can reduce the vibration up to 20%, which is transmitted to the palm (Hewitt et al. 2015). During the comparison of bare hand drilling with nylon, rubber and leather gloves, the average total vibration magnitude (a_{hv}) was reduced by 16.78%, 30.91% and 23.78%, respectively. Each glove type showed a significant reduction in total vibration.

To ensure subject/operator safety and reduce the likelihood of HAV syndromes, the use of rubber gloves is highly recommended, with ongoing monitoring and adjustments based on specific A(8) values and applicable standards. The rubber gloves reduce HTV exposures more than leather gloves when using handheld vibrating tools (Chaturvedi et al. 2012). The glove used for drilling is designed for the reduction of hand fatigue, which provides the required grip strength and manual dexterity more than leather gloves. The contact between the hand and tool handle might be separated due to the large vibration exposure. The reduction was 12.84%, 35.85%, and 22.36% for nylon, rubber and leather when compared with barehand respectively.

Electric impact drills may produce vibrations same as stone hammers and rock drills. In such cases, the tool produces a high-frequency range. In such conditions, the glove decreased the vibration transferred to the palm by over 30% (Hewitt et al. 2015). Therefore, wearing nylon gloves causes a noticeable reduction in the maximum amplitude in the frequency spectrum as compared to the barehand condition. This indicates that vibrations are transferred to the hand and are somewhat dampened by nylon gloves. The decrease is less noticeable than when rubber gloves are used. Rubber gloves are excellent at absorbing vibrations, as evidenced by this significant amplitude reduction. Rubber gloves are useful in reducing vibrations transmitted by the hands, particularly at higher frequencies. Leather gloves have a level of

damping effect that is intermidiate between that of rubber and nylon gloves. From the frequency spectrum analysis of HTV, it is evident that different types of gloves offer varying levels of rms acceleration. The highest vibration amplitudes are seen with barehand exposure, which suggests no damping effect. These results imply that the unique vibration exposure conditions and the requirement for dexterity should be taken into consideration when selecting the glove material. Rubber gloves are advised for work involving much vibration or while drilling harder materials. Leather gloves are appropriate for moderate vibration and require a certain amount of flexibility. When maximum dexterity is desired, low to moderate vibration intensities can be used with nylon gloves. The observed directional differences in RMS acceleration values at low frequencies can be attributed to the anisotropic properties of the hand and glove materials, the characteristics of the vibration source, and the measurement setup. By ensuring the best possible protection and performance, this selection procedure improves worker comfort and safety. However, usage of a vibration-reducing glove could increase the vibration, especially in the fundamental vibration frequency range and generally reduce the impedance of the hand-arm system applied to the tool (Dong et al. 2009) the tool (Dong et al. 2009). This allows the frequency-weighted acceleration of the tool to be increased when the glove is used, thereby reducing the effectiveness of the glove. There might be more effects on the human hand due to the factors that favor reducing tool vibration. Individuals/operators can give greater force on the drill machine to control the increase in tool vibration while using the gloves. This results in a great reduction in hand sensation, perception, and grip force. According to (McDowell et al., 2013), in certain frequency ranges, an increase in hand force results in an increase in glove efficacy. However, an increase in hand forces is considered as the adversarial effect of the glove; the reason behind this is that an increase in hand forces causes injuries more than a reduction in vibration exposure.

Limitations

There are quite a few limitations in the current study. The study did not consider variations in the feed force during drilling and the posture of the participants. All the participants in the study were advised to drill keeping their arm 90° in the sagittal plane and the drilling machine perpendicular to the drilling material. The study also involved using a soft material (teak wood). By using harder materials, the a_{hv} and A(8) values are likely to vary based on the material hardness. Also, the feed force was also approximately kept constant. The thickness of the glove material (2 mm) was also kept constant in this study. Thicker glove material can minimize the damping effect of vibrations. Regular monitoring and evaluation of HTV exposure and the implementation of safety measures such as wearing gloves can prevent HAVS among employees working in different industries. The vibration effects can be minimized if proper training and skills are followed by the operator. Fixing the drill bit into the drilling machine properly, using the correct drill bit (HSS or carbide drill bits for semi-hard and harder materials), rotation of workers to minimize A(8), and proper posture of the operator's arm during drilling are a few of the protocols to be followed to prevent HAVS.

Conclusion

A low-cost, portable and simple vibration measuring system (for HTV) was designed and developed. The study involved in comparing the usage of different protective gloves to reduce the exposure of the vibration that is transmitted to the hand during the drilling process. The average percentage reduction of total vibration magnitude (a_{hv}) by nylon, rubber, and leather gloves was 16.78%, 30.91%, and 23.78%, respectively, when compared to barehand. The most significant vibration amplitudes occurred at 10 and 100 Hz. The A(8) or exposure action value (EAV) has shown a considerable reduction of 35.85% while using a rubber glove in comparison with barehand. For the drilling operation, individuals wearing protection gloves have a lower risk of HTV syndrome, where the rubber is the most effective at reducing vibration exposure, which significantly lowers the A(8) than EAV values, indicating a lower risk of HAV syndrome. The

developed device can be implemented in real-time operations as it can be integrated into worker's equipment used in outdoor operations.

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